INTRODUCTION

This instruction outlines the field engineering procedures to be followed when inspecting MS6001B Gas Turbine Load Gear.

The inspection procedures provided in this article cover the general condition of the gear teeth, gear tooth contact between the pinion and bull gear, cleanliness inside of the gear housing and correct flow pattern of lube oil nozzles.

DESIGN INFORMATION

The Load Gear provided with the MS 6001 Gas Turbine is an up-over, double helical, high speed design which places the high speed pinion gear shaft (that connects to the MS 6001B Gas Turbine) above the lower speed bull gear (that connects to the generator). (See Fig. 1).

This procedure is based on the Philadelphia Gear S674-A gear assembly. (Philadelphia drawing 03031-2457). This procedure is also applicable to Westech load gear assemblies that have been provided for some MS 6001 Gas Turbine units.

INSPECTION RECOMMENDATIONS

The load gear inspection should be performed during a gas turbine maintenance inspection. Whenever possible, the generator coupling should remain bolted to the bull gear during the inspection to add loading on the teeth, thereby improving the accuracy of the contact check.

When the contact inspection is found to be less than 75 percent (of the face area), a second contact check is recommended to verify that the findings of the first check are accurate. If the contact is still found to be less then 75 percent, the vendor should be contacted for their evaluation and recommendations.

SPECIAL TOOLS, FIXTURES AND MATERIALS

The special tools and fixtures furnished by General Electric during the initial installation will be required for load gear inspection. These include the field fabricated gear turning tool (or coupling wrench), and the alignment centering ring provided by the manufacturer (when re-alignment is necessary).

The special materials required to conduct a gear contact inspection include a tube of Prussian Blue or Dykm Blue (not Layout Blue which dries), a tube of Permanent Artist Oil Yellow Paint – Cadmium Yellow Medium, and a pint of parts cleaner such as alcohol or red dye check liquid penetrant cleaner. Diesel oil, kerosene, and similar oily fluids do not produce satisfactory results because the leave oily residue.

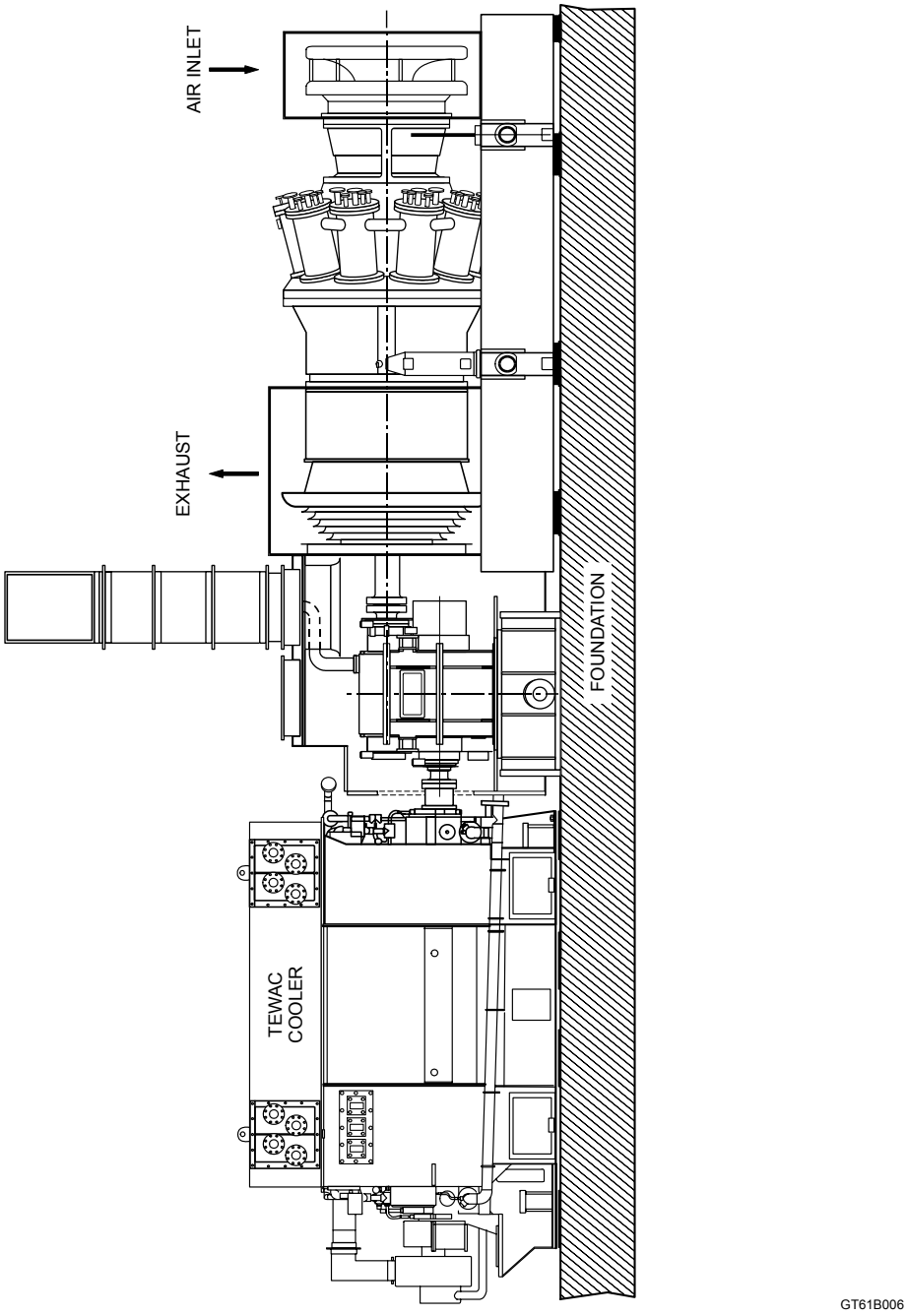


Figure 1
MS 6001B Gas Turbine w/DLN-1 Combustion, S 674 Load Gear & 6A6 Generator – Typical

REFERENCE DOCUMENTS

No.	Description
HSM-175	Philadelphia Load Gear Service Manual
TH511, TH514	Internal Alignment of Ship Propulsion Gears – Procedures for Tooth Contacting
EM 5316	MS 61B (Lucas Aerospace) Load Coupling Installation
EM 5211	MS 61B Load Gear Field Inspection – Main Bearing and Gear Shaft Journals

Method of Inspection

The following is the recommended method for inspecting the MS 61B load gear pinion and bull gear shafts tooth contact–mesh alignment.

1. Insure that the turbine-generator and auxiliaries are properly tagged and locked-out.
2. Remove the inspection cover from the left upper side of the load gear case (some newer gear cases may have the inspection cover on the right hand side.) (See Figs. 2 and 3).

NOTE: Prior to checking the contact, the direction of rotation and active (driving) tooth faces of the driving gear (pinion) must be determined. (See Fig. 4)

3. Select five or six teeth on the active meshing faces of both the pinion and bull gears, and thoroughly clean the teeth areas with a parts cleaner. (See Figs. 5, 6 and 7–Detail “A”).
4. When all dirt, grease and oil are removed, apply a very thin coating of yellow paint to four or five of the pinion gear shaft teeth on both helices. The coating must be applied with a soft cloth, not a brush. (See Figs. 8 and 9–Detail “B”).
5. A very thin coating of Prussian Blue or Dykm Blue is then applied to four or five of the bull gear shaft (driven rotor) teeth on both helices using a soft cloth, not a brush. (See Figs. 8 and 9–Detail “B”).

NOTE: It is essential that the coatings be applied evenly along each helix and not exceed 0.001 inches (approximate) in thickness. An excessive thickness of coating will result in a contact pattern with little or no value.

6. Rotate the gear set through mesh, by jogging the starting means torque converter. **Do not roll the gears more then once through each application of colors.**

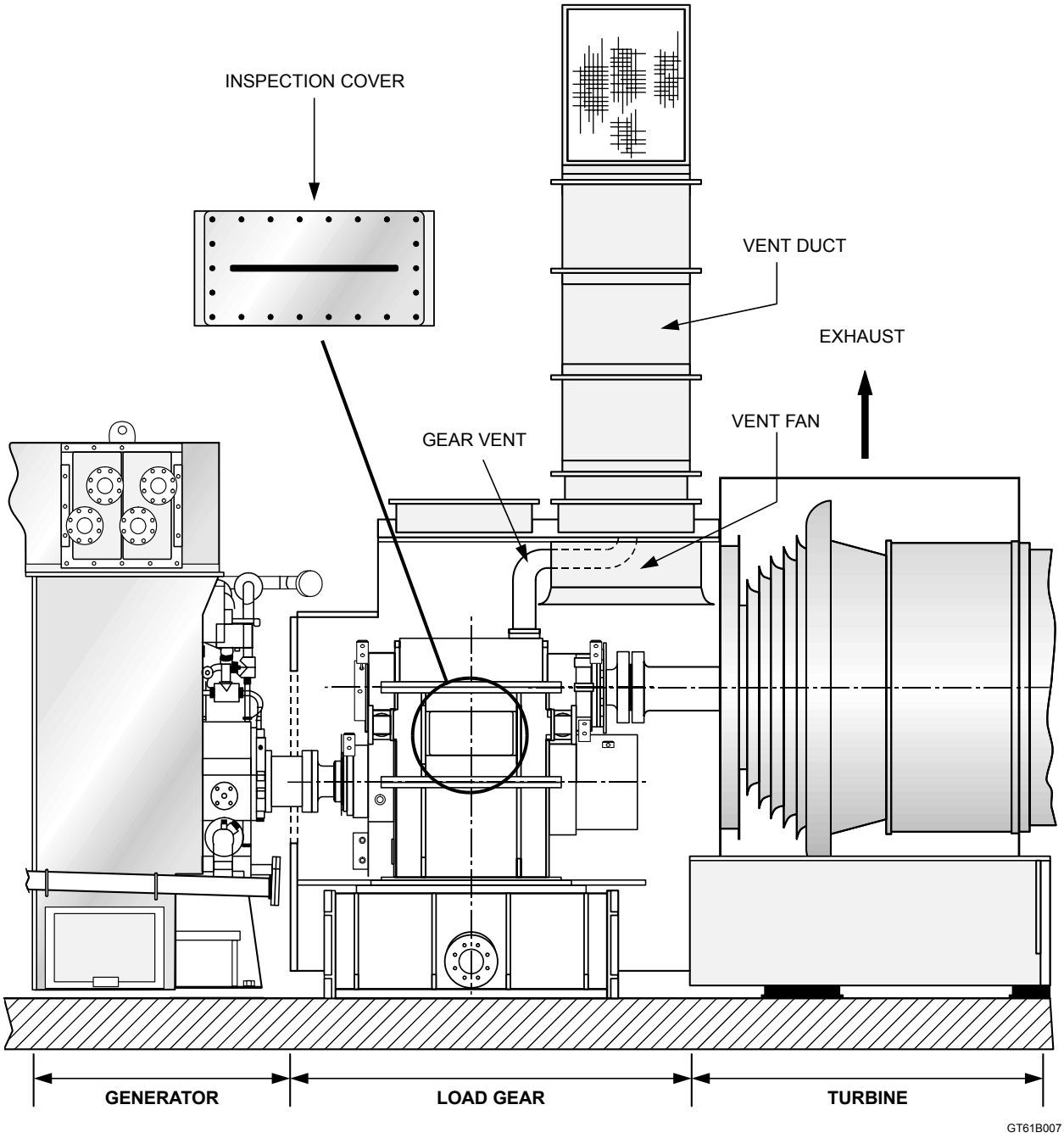
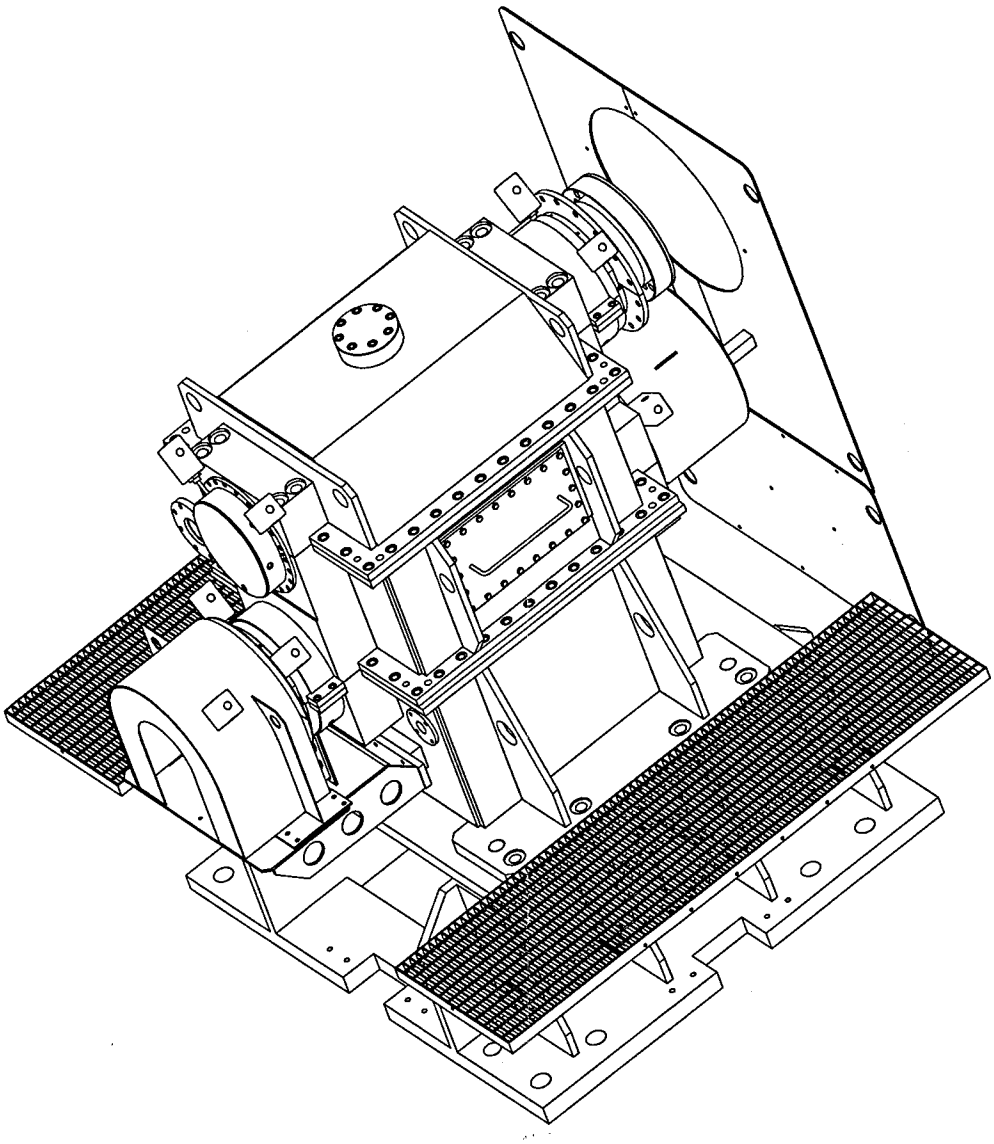


Figure 2. Philadelphia S–674 Load Gear



GT61B008

Figure 3. Philadelphia S–674 Load Gear — Typical Layout (Plan View)

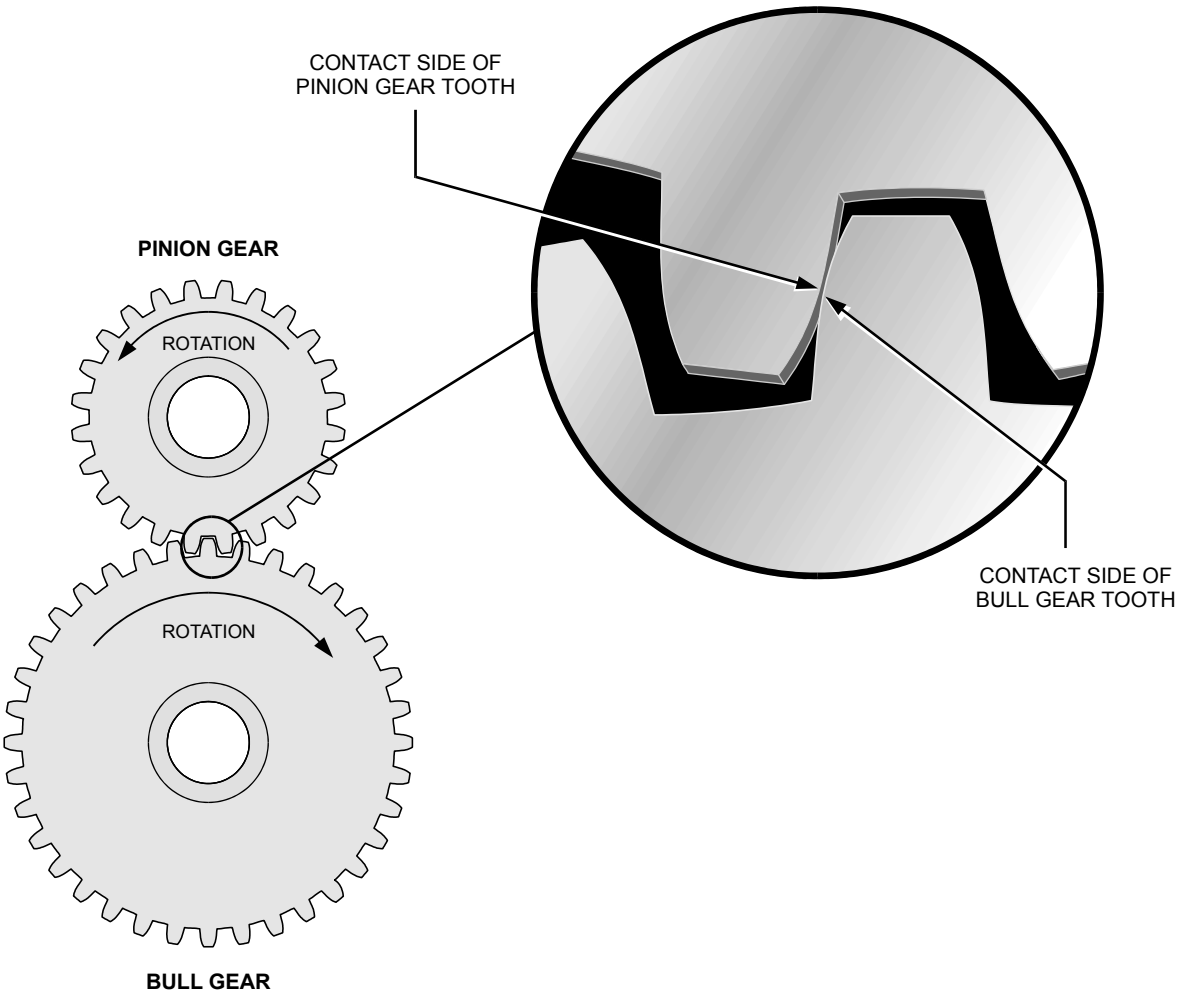


Figure 4. Gear Rotation and Tooth Action Diagram

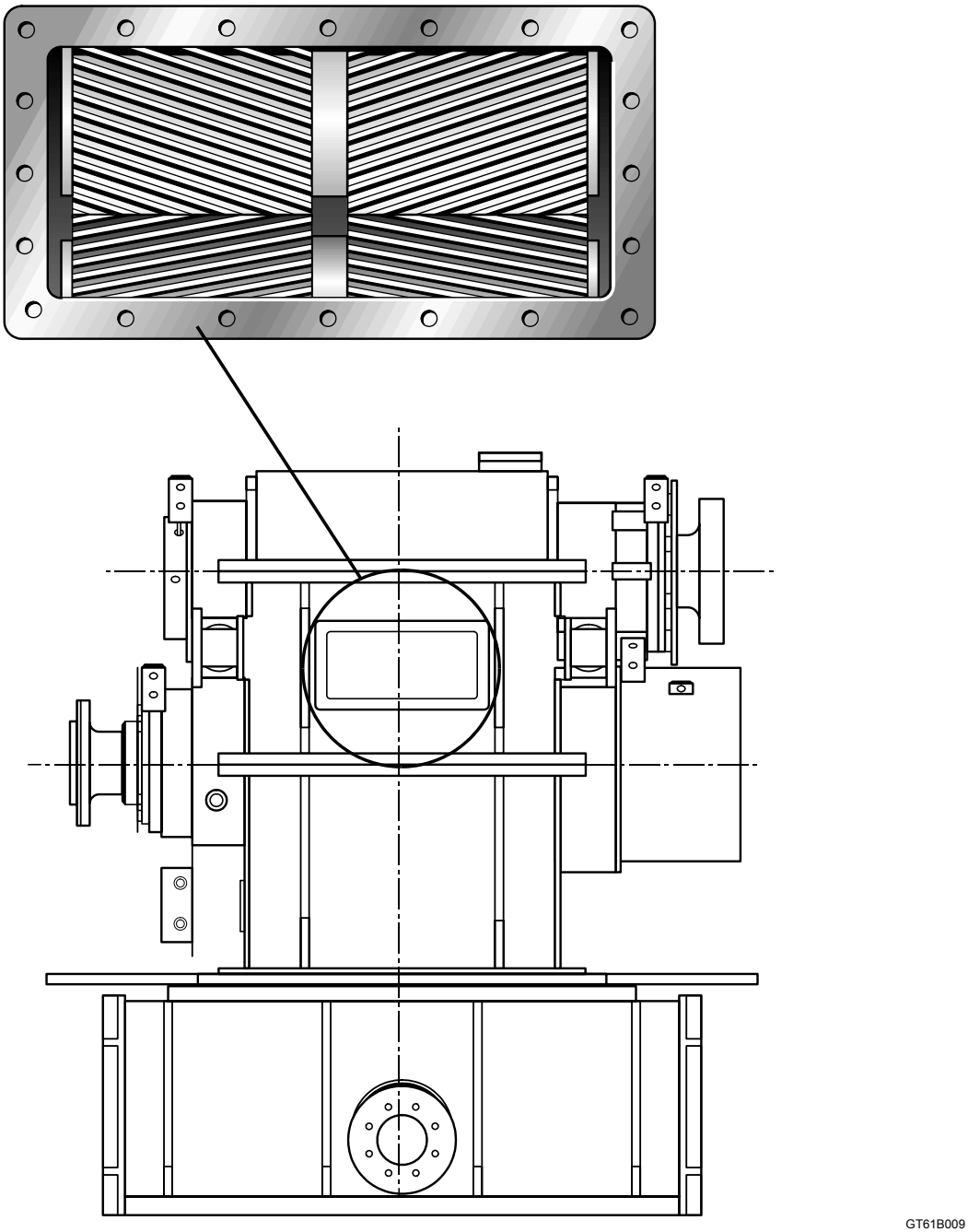


Figure 5. Bull and Pinion Gear with Inspection Cover Removed

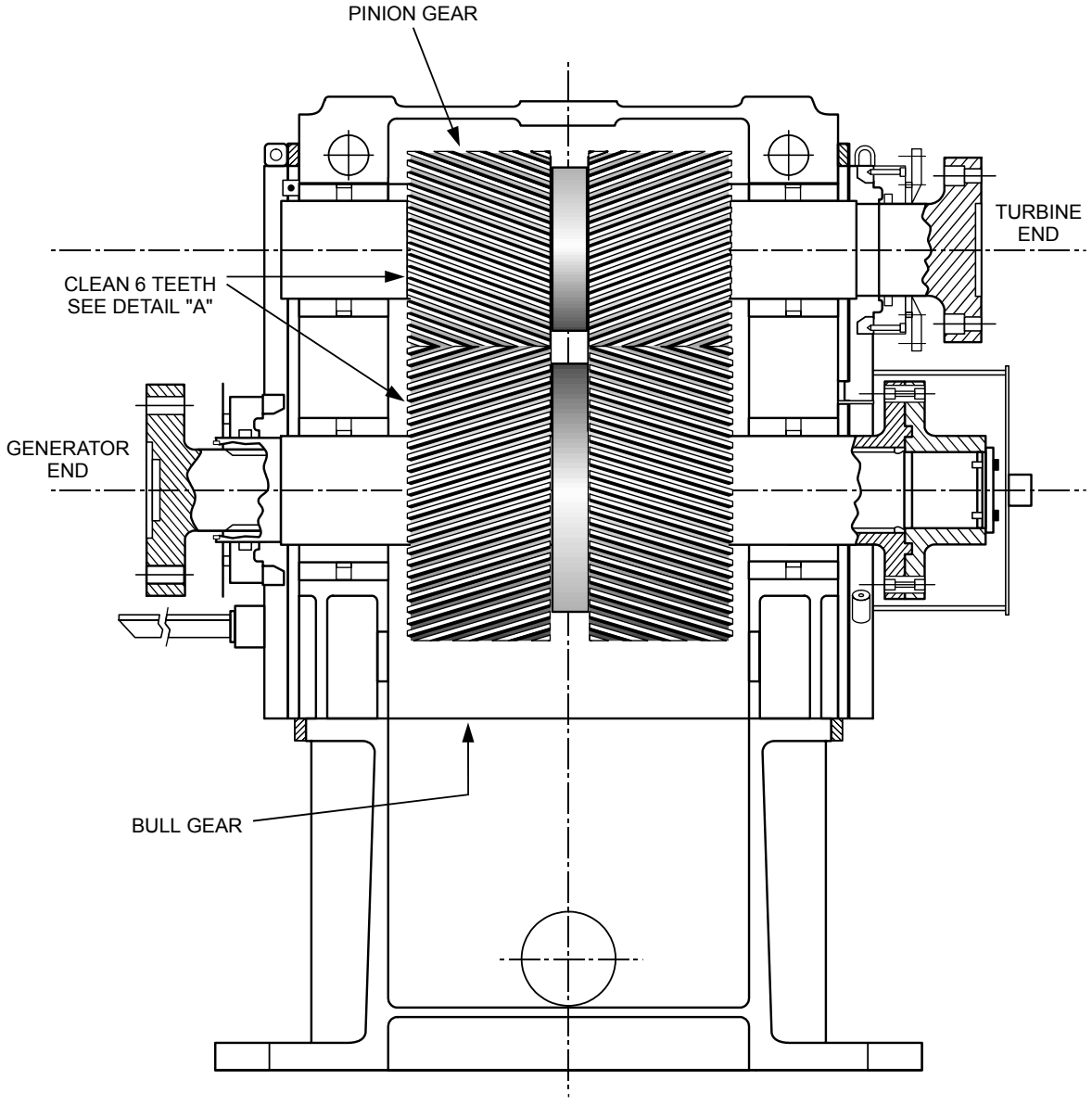


Figure 6. Bull and Pinion Gear Tooth Contact Check Area

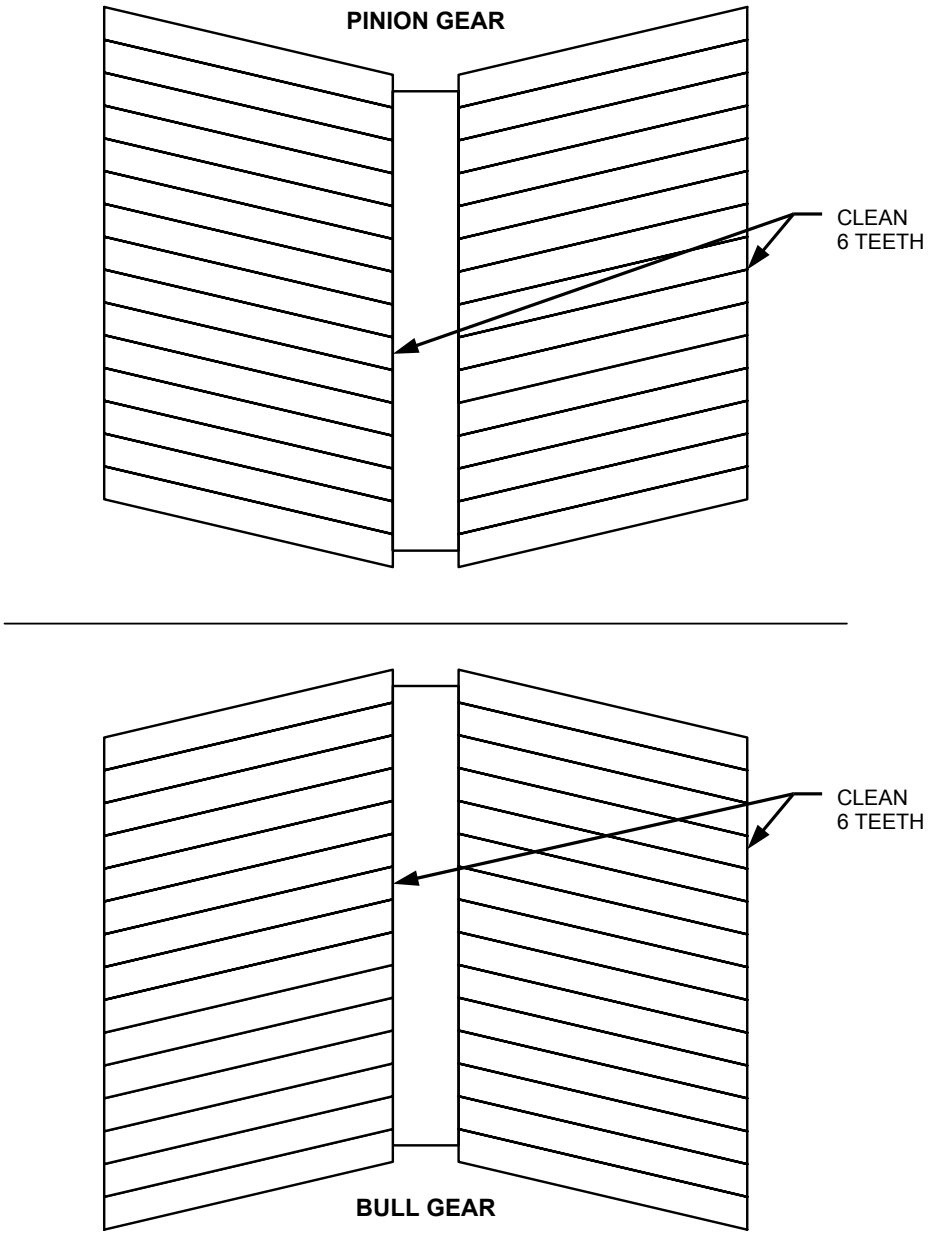


Figure 7. Detail "A" (See Fig. 6)

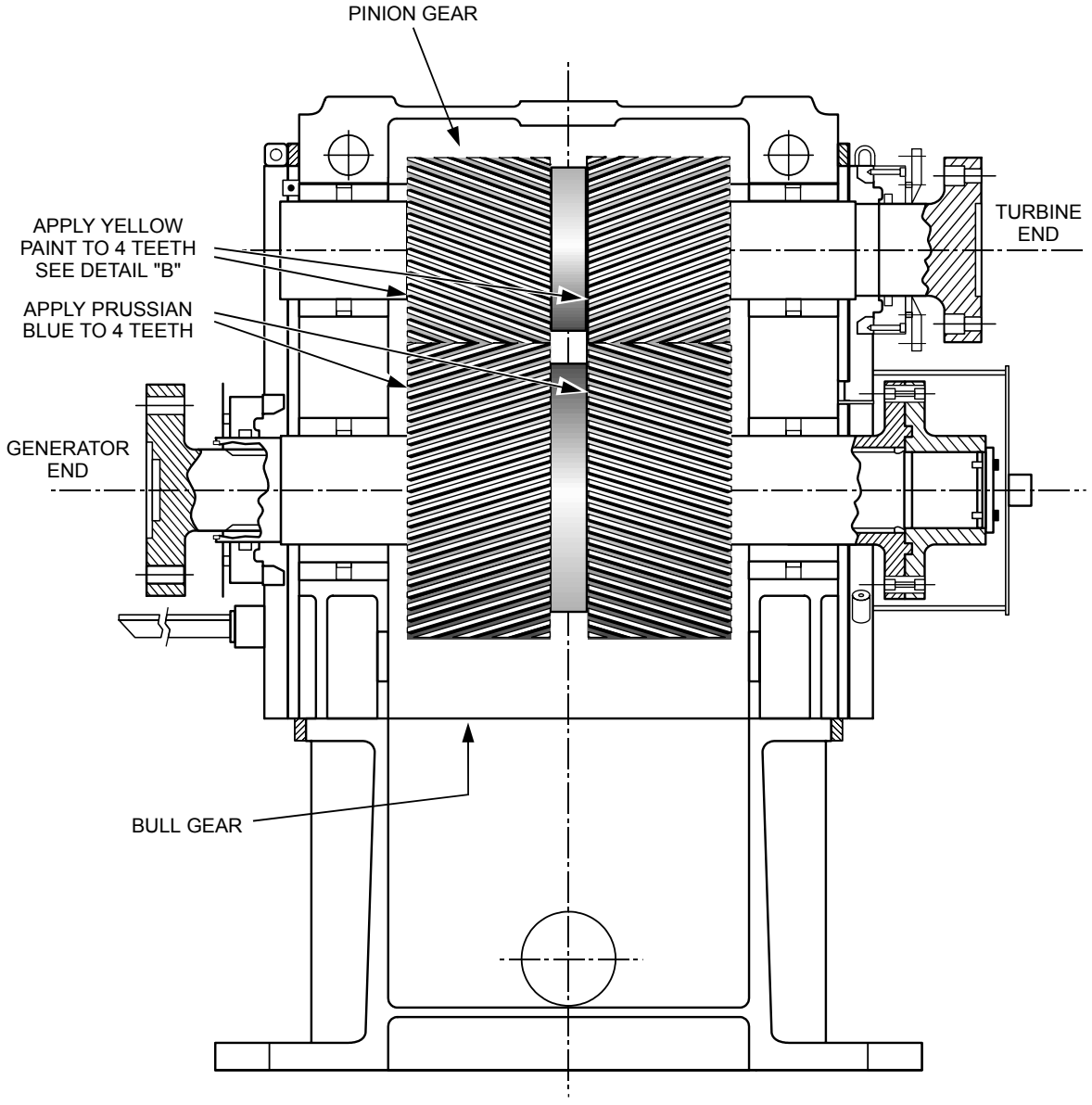


Figure 8. Paint Application Areas

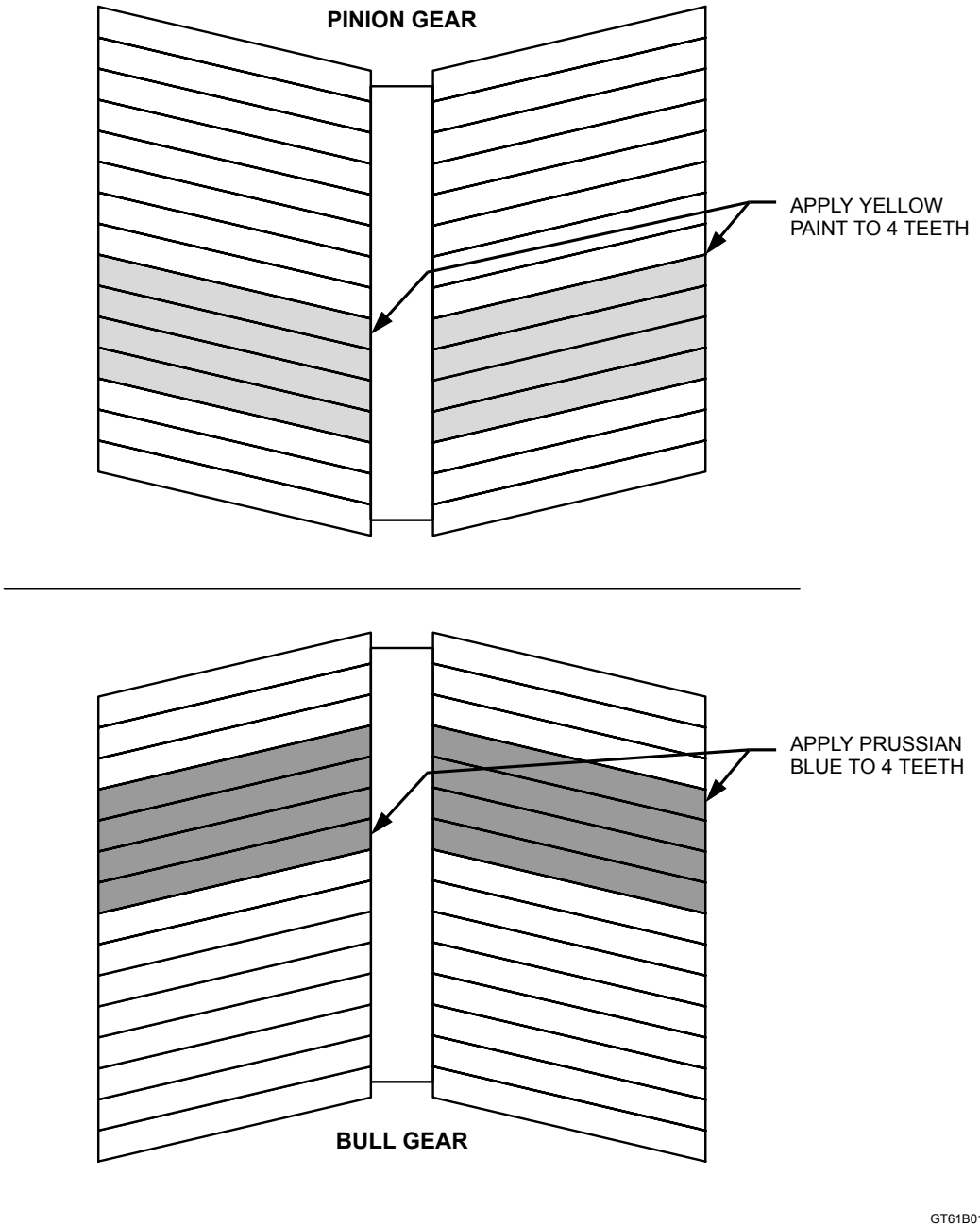


Figure 9. Detail "B" (See Fig. 8)

7. Analyze the alignment according to the following criteria:

   A. If the gear shaft internal alignment is correct, the transfer of yellow onto the blue will be even and full across both helices. (See Fig. 10).

   B. If the transfer of yellow onto the bull gear is not uniformly distributed, there are three basic patterns which indicate gear misalignment that is near the limit of normal operation. (See Fig. 11, Sketches A, B and C):

      Sketch A Heavy yellow (or contact) at the base of the helix and at the apex of the other, indicates internal misalignment. This is the most common finding.

      Sketch B Heavy yellow (or contact) at the base of both helices indicates very slight difference in the helix angles between the pinion and bull gears.

      Sketch C Heavy yellow (or contact) at the apex of both helices indicates a very slight difference in the helix angles between the pinion and bull gears.

      Sketch D Under full load, the yellow (contact) for each of the three misalignment conditions (shown in sketches A, B and C) will be more evenly distributed across the helices but the contact will not be full or completely even as a normal transfer contact pattern. (See Fig. 10).

      NOTES: No correction of misalignment conditions are possible in the field. The helix angle difference indicated in these examples is extremely slight, and correction by re-cutting is not considered to be necessary.

      The distribution of paint contact indicated in these sketches does not apply to high ratio gearing.

Record Contact Inspection

The contact check and inspection should be documented as follows:

1. Photograph the gear contact checks, to provide records for comparison with future inspections.
2. Redraw the Gear inspection Report Form as required to accept the tape strip records. (See Fig. 12).
3. Carefully place transparent tape strips (1 to 1-1/2 inches wide) over the contact pattern left on each active tooth face, and gently press to transfer the pattern to the tape strip. Slowly lift each tape and apply it the report form.
4. Complete the gear tooth condition reports. (See Figs. 12 and 13A and 13B).

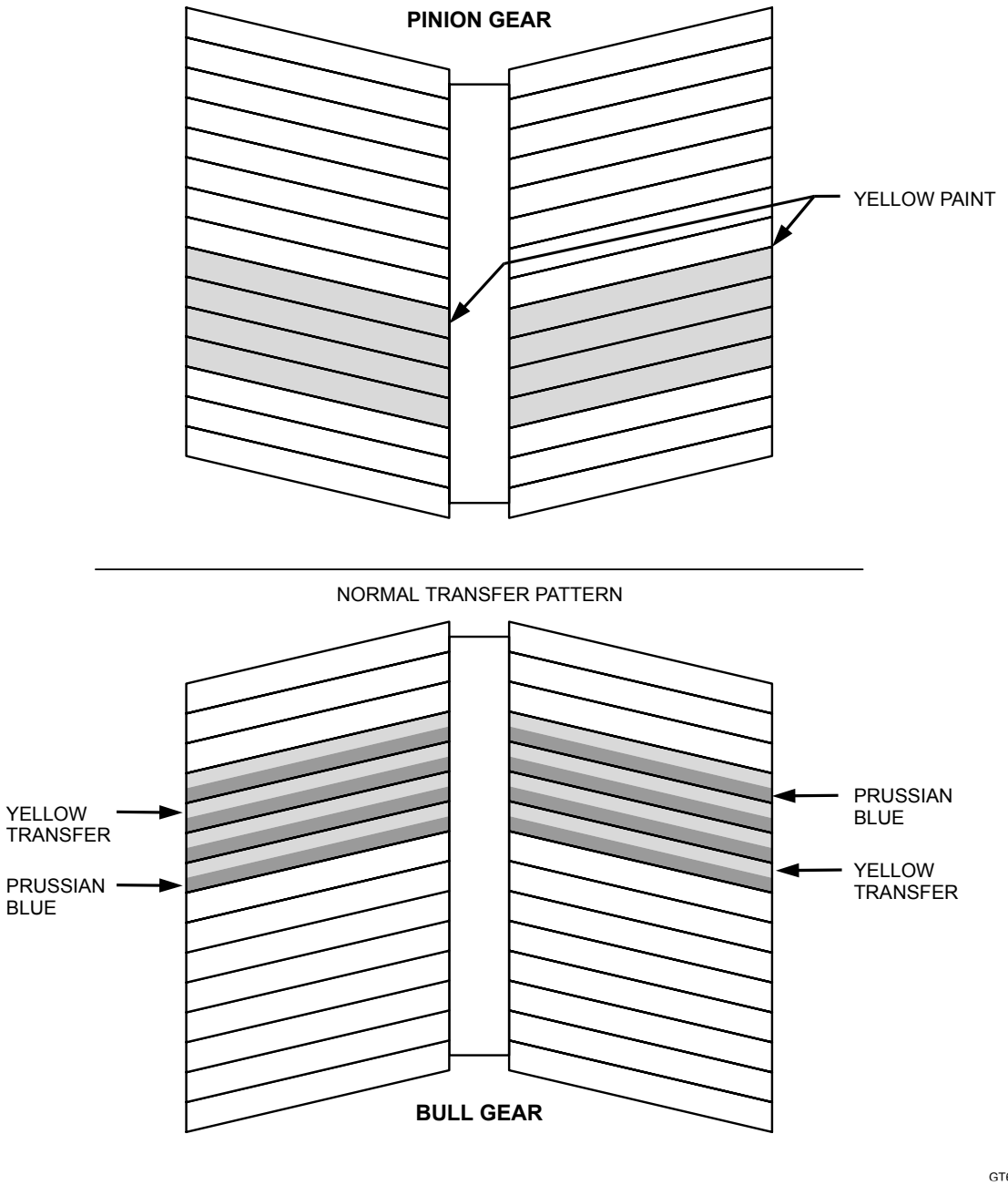


Figure 10. Normal Tooth Contact Transfer Pattern

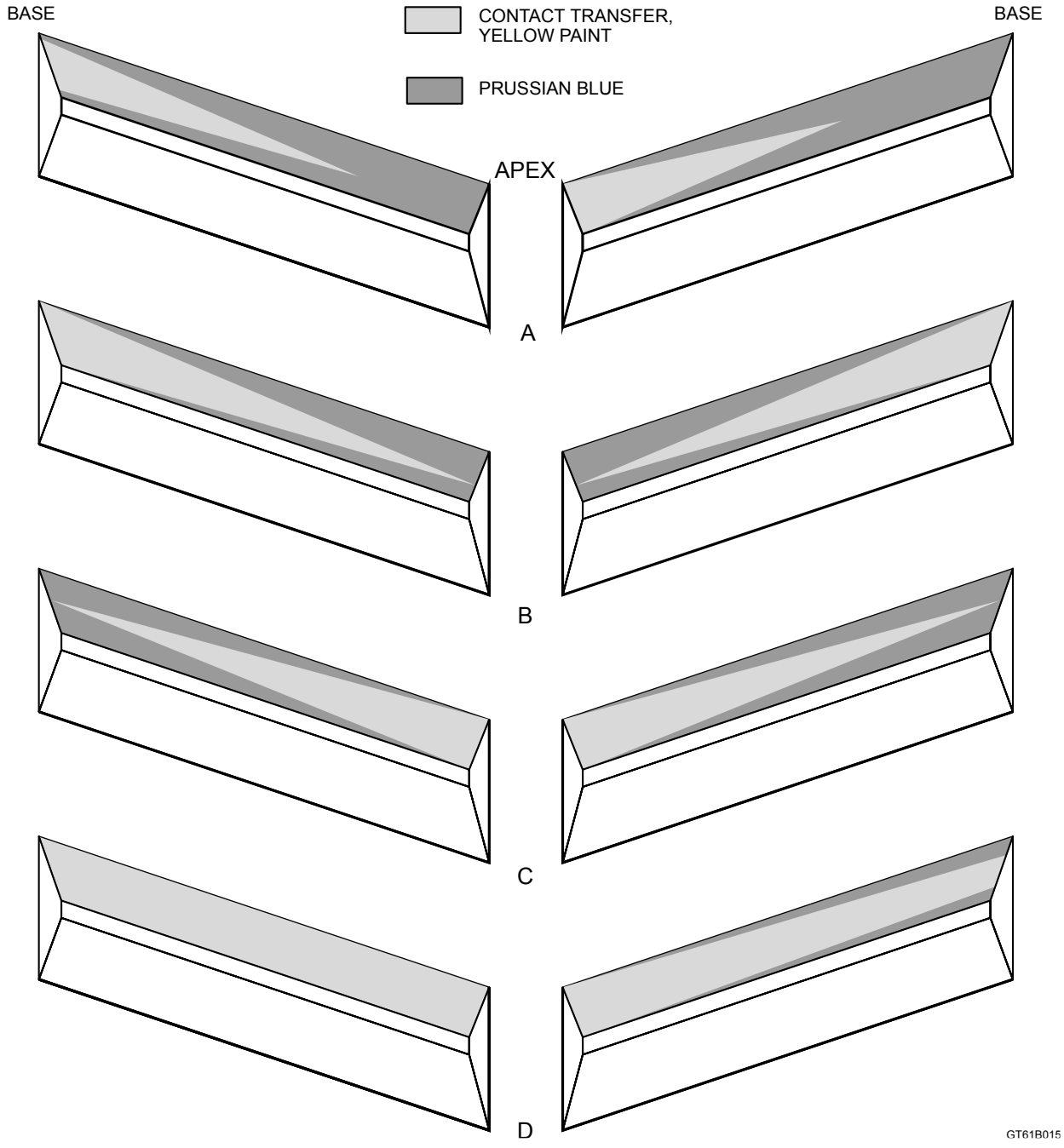


Figure 11. Gear Misalignment Contact Patterns

REASSEMBLE INSPECTION COVER

1. Using a parts cleaner, thoroughly clean the colorings applied to the teeth of the active meshing faces of the rotors.
2. Apply a bead of red RTV to the inspection cover (or install a new gasket where used), and reassemble to the left (or right) upper side of the load gear.
3. Make up cover bolting. Refer to the vendor service manual for torque values.

Note: The gear case bolt holes should be filled with red RTV. The purpose is to fill the void around the bolts and seal the bolt heads, preventing oil leagage from the casing flang through the bolt holes, which may occure after load gear inspection.

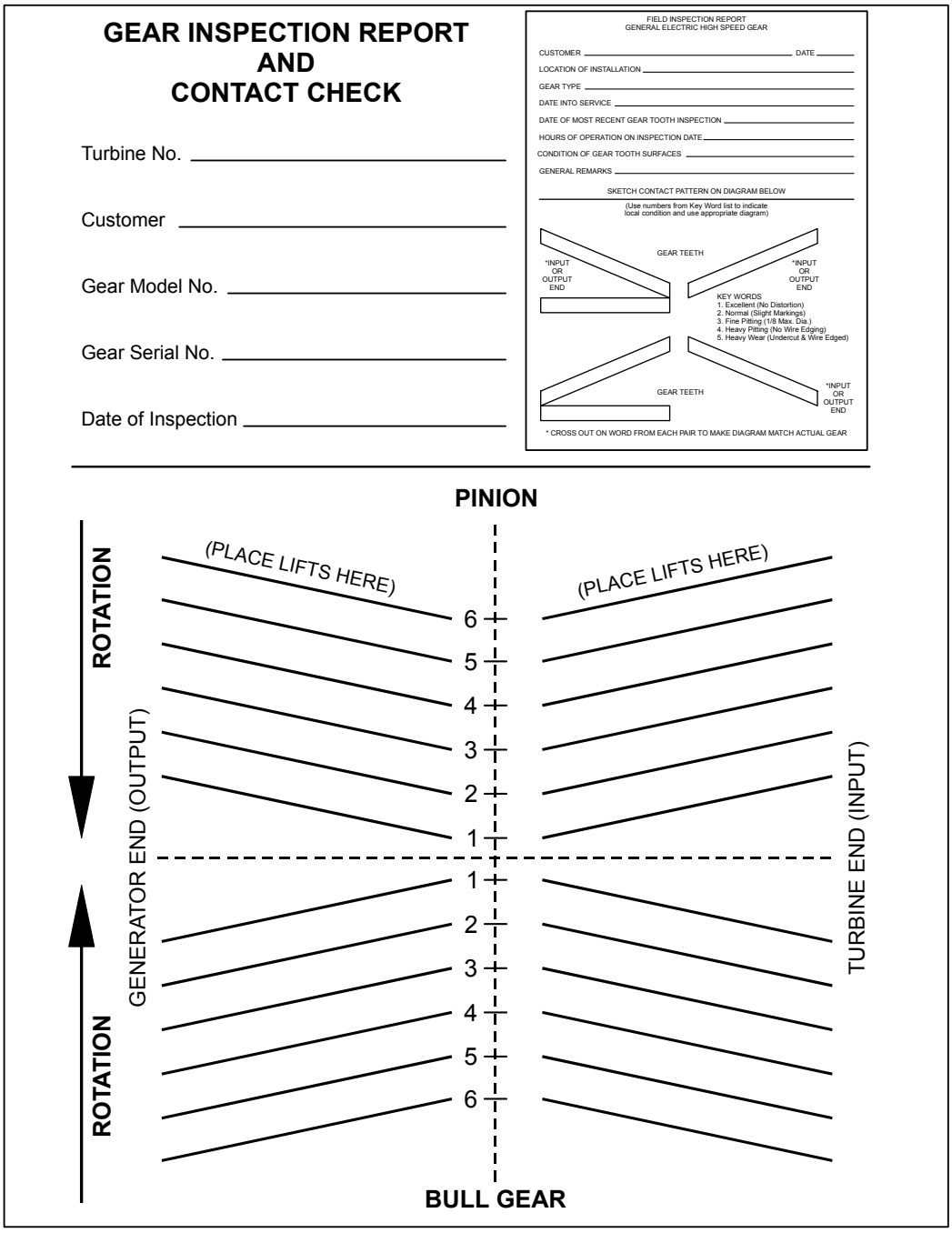


Figure 12. Gear Inspection Report and Contact Check — Example of Lift Form

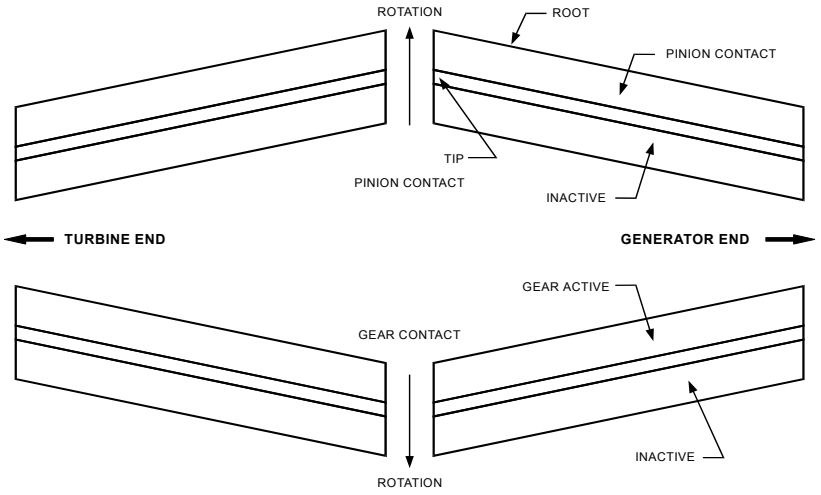
INSPECTION REPORT
Gas Turbine Maintenance

Gear Maintenance

MS6001B ***Load Gear***

Date _____ Turbine S/N _____ Prepared By _____
FSR # _____ Generator S/N _____ Photos Enclosed? _____
Data Type _____ Gear Model _____ Sketches Enclosed? _____

Contact Pattern Chart



GT61B017

Date _____

Condition of Tooth Surfaces (Check below)

- ☐ Excellent (No Distress)
- ☐ Normal (Slight Marking)
- ☐ Fine Pitting (1/16" Max. Dia.)
- ☐ Heavy Pitting (Over 1/16" Dia.)
- ☐ Slight Wear (No Wire Edging)
- ☐ Heavy Wear (Undercut & Wire Edged)

- Oil Feed Orifice Size _____
- Oil Spray Pattern
  - ☐ Safisfactory ☐ Unsatisfactory
- Indication of Babbitt Distress at Bearings
  - ☐ Yes ☐ No
- Loose Debris in Bottom of Gear Box
  - ☐ Yes ☐ No

Comments:

GT61B017a

Figure 13–A. Contact Pattern Inspection Form — Example

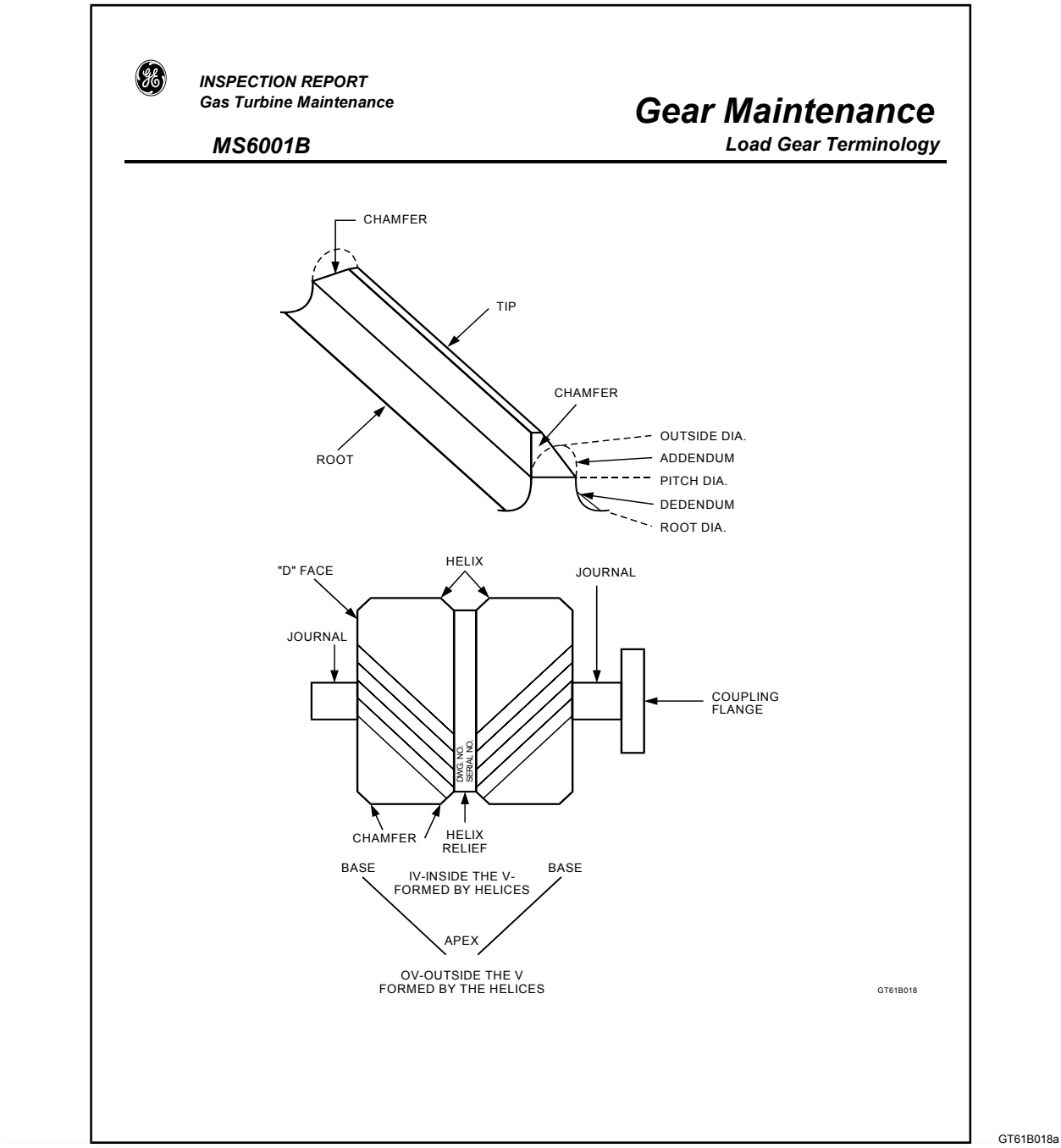


Figure 13–B. Load Gear Terminology